

Mandatory exercise 3

The deadline for this mandatory exercise is **Tuesday October 24 at 23:59**. Do the following exercises.

1. Consider the basis for P_2 given by $B = \{p, q, r\}$, where

$$p = 3x^2 - 2$$

$$q = 2x^2 + x$$

$$r = x - 1$$

Compute the coordinates of the polynomial $s = 10x^2 + x - 10$ relative to B .

2. Find the linear combination of the functions

$$f(x) = 2x^2 + x$$

$$g(x) = x + 1$$

that best fits the observation points $\{(-1, -2), (0, 2), (1, 5)\}$. In other words, find the numbers a, b , such that $h(x) = a \cdot f(x) + b \cdot g(x)$ minimises the sum

$$(h(-1) + 2)^2 + (h(0) - 2)^2 + (h(1) - 5)^2$$

Explain your reasoning. The object of this exercise is not just to compute the numbers a and b , but also to write a convincing argument that these numbers indeed minimise the sum above. To do this, follow the procedure in the notes: construct the set V of all tuples of the form $(h(-1), h(0), h(1))$, find a basis for this as a subspace of $\mathbb{R}^3$, and compute the a, b as coordinates of some projection down to V .

1. Consider the basis for P_2 given by $B = \{p, q, r\}$, where

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Compute the coordinates of the polynomial $s = 10x^2 + x - 10$ relative to B .

Base is given by p, q, r

$$B = \{(3x^2 - 2), (2x^2 + x), (x - 1)\}$$

$$s = 10x^2 + x - 10.$$

find a, b, c such. $s = a \cdot p + b \cdot q + c \cdot r$

$$10x^2 + x - 10 = a(3x^2 - 2) + b(2x^2 + x) + c(x - 1)$$

$$10x^2 + x - 10 = 3x^2a - 2a + 2x^2b + xc + cx - c$$

System of Lin. eq.

$$\begin{array}{l} \text{I. } 3a + 2b = 10 \\ \text{II. } b + c = 1 \\ \text{III. } -2a - c = -10 \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|c} 3 & 2 & 0 & 10 \\ 0 & 1 & 1 & 1 \\ 2 & 0 & -1 & -10 \end{array} \right]$$

from I. system
get a

$$3a = 10 - 2b$$

$$a = \frac{10 - 2b}{3}$$

$c = 1 - b$ from II. system get c

now plug a, c into III.

$$-2\left(\frac{10 - 2b}{3}\right) - (1 - b) = -10$$

$$\frac{-20 + 4b}{3} - 1 + b = -10$$

$$\frac{-20+4b-3+3b}{3} = -10 \quad \backslash \cdot 3$$

$$-20+4b-3+3b = -30$$

$$7b = -30+20+3$$

$$7b = -7$$

$$b = -1$$

$$\text{II. } b+c = 1$$

$$-1+c = 1$$

$$c = 2$$

$$\text{III. } -2a - c = -10$$

$$-2a = -10 + c$$

$$-2a = -10 + 2$$

$$-2a = -8$$

$$a = 4$$

So the coordinates of s , relative to B are $a, b, c \rightarrow (4, -1, 2)$

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Explain your reasoning. The object of this exercise is not just to compute the numbers a and b , but also to write a convincing argument that these numbers indeed minimise the sum above. To do this, follow the procedure in the notes: construct the set V of all tuples of the form $(h(-1), h(0), h(1))$, find a basis for this as a subspace of $\mathbb{R}^3$, and compute the a, b as coordinates of some projection down to V .

points: $(-1, -2), (0, 2), (1, 5)$

$$h(x) = a \cdot f(x) + b \cdot g(x)$$

$$h(x) = a \cdot (2x^2 + x) + b \cdot (x + 1)$$

$$h(x) = 2x^2 a + xa + bx + b$$

$$h(-1) = 2a - a - b + b$$

$$h(-1) = a$$

$$h(0) = b$$

$$h(1) = 2a + a + b + b$$

$$h(1) = 3a + 2b$$

$$V = \{ (h(-1), h(0), h(1)) \mid f(h) = a \cdot f(x) + b \cdot g(x) \}$$

$$V = \{ a, b, 3a+2b \mid a, b \in \mathbb{R} \}$$

→ find lin. combinations to get $a, b, 3a+2b$

$$\begin{matrix} a(1 & 0 & 3) = r_1 \\ b(0 & 1 & 2) = r_2 \end{matrix} \rightarrow \text{because } (1a + 0a + 3a) + (0b + 1b + 2b) = a, b, 3a+2b$$

$$B = \left\{ \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$$

→ these vectors describe set V with minimal vectors (Basis)

If Basis is known, I can write matrix A

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \quad \vec{m} = \text{Y-cords of points} \\ \vec{m} = (-2, 2, 5)$$

$$[P]_B = (A^T \cdot A)^{-1} \cdot A^T \cdot \vec{m} \quad \text{need } A^T \cdot A, (A^T \cdot A)^{-1}$$

$$A^T \cdot A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1+0+9 & 0+0+6 \\ 0+0+6 & 0+1+4 \end{bmatrix}$$

$$A^T \cdot A = \begin{bmatrix} 10 & 6 \\ 6 & 5 \end{bmatrix}$$

$$(A^T \cdot A)^{-1} = \frac{1}{\det(A^T \cdot A)} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\rightarrow a \cdot d - b \cdot c$$

$$(A^T \cdot A)^{-1} = \frac{1}{50-36} \cdot \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix}$$

$$(A^T \cdot A)^{-1} = \frac{1}{14} \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix}_B = \frac{1}{14} \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix}_B = \frac{1}{14} \cdot \begin{bmatrix} 5 & -6 & 3 \\ -6 & 10 & 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix}_B = \frac{1}{14} \cdot \begin{bmatrix} -10 & -12 & 15 \\ 12 & 20 & 10 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix}_B = \frac{1}{14} \cdot \begin{bmatrix} -7 \\ 42 \end{bmatrix}$$

$$\downarrow \quad \downarrow$$

$$a = \frac{-7}{14} \quad b = \frac{42}{14}$$

$$\boxed{a = -\frac{1}{2}} \quad \boxed{b = 3}$$

Explanation:

$f(x)$ → QUADRATIC FUNCTION

$g(x)$ → LINEAR FUNCTION

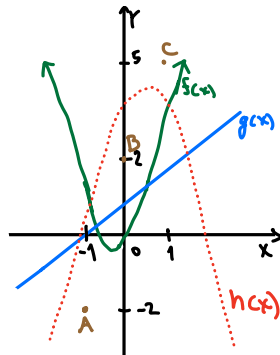
Points: $A = [-1, -2]$

$B = [0, 2]$

$C = [1, 5]$

$h(x)$ → expected function, such that distance between

Point and function is shortest



$$h(x) = a \cdot f(x) + b \cdot g(x)$$

this means that function $h(x)$ is lin combination of $f(x)$ and $g(x)$ with parameters (a, b)

Such that they minimize the sum

$$(h(-1)+2)^2 + (h(0)-2)^2 + (h(1)-5)^2 = \text{Sum of dist.}$$

So the sum of the distances [Point and $h(x)$] is Shortest

$$\text{distance for point B: } (h(-1)+2)^2$$

B: ...

C: ...

so choosing right parameters a, b I will get function $h(x)$ such that SUM of the distances between $h(x)$ and points will be shortest.

